

CURRICULUM VITAE – PERCY LIANG

Personal Details

Name: Percy Liang
Email: плианг@cs.stanford.edu

Education

2011 Ph.D. in Computer Science, University of California at Berkeley
2005 M.Eng in Computer Science, Massachusetts Institute of Technology
2004 B.S. in Computer Science, Massachusetts Institute of Technology

Experience

2019– Associate Professor at Stanford University
2012–2019 Assistant Professor at Stanford University
2011–2012 Post-doc at Google, New York, NY
2010 Intern at Intel Labs, Berkeley, CA
2005 Intern at Microsoft Research, Cambridge, UK
2004 Intern at Microsoft Research, Redmond, WA
2002 Intern at Google, Mountain View, CA

Awards and Honors

2019 Presidential Early Career Award for Scientists and Engineers (PECASE)
2018 Best short paper award (ACL)
2017 Best paper award (ICML)
2017 Best paper award (COLT)
2016 IJCAI Computers and Thought Award
2016 NSF CAREER Award
2015 Sloan Research Fellowship
2014 Microsoft Research Faculty Fellowship

Publications

BOOK CHAPTERS

1. Percy Liang, Michael I. Jordan, and Dan Klein. Probabilistic grammars and hierarchical Dirichlet processes. In *The Oxford Handbook of Applied Bayesian Analysis*. 2009.
2. Stefan Wager, Will Fithian, and Percy Liang. Data augmentation via Lévy processes. In *Perturbations, Optimization and Statistics*. 2016.

REFEREED JOURNAL PUBLICATIONS

1. Percy Liang and Christopher Potts. Bringing machine learning and compositional semantics together. *Annual Reviews of Linguistics*, 1(1):355–376, 2015.
2. Percy Liang, Michael Jordan, and Dan Klein. Learning dependency-based compositional semantics. *Computational Linguistics*, 39:389–446, 2013.

1. John Thickstun, D. Hall, Chris Donahue, and Percy Liang. Anticipatory music transformer. *Transactions of Machine Learning Research (TMLR)*, 0, 2024.
2. Rohith Kudritipudi, John Thickstun, Tatsunori Hashimoto, and Percy Liang. Robust distortion-free watermarks for language models. *Transactions of Machine Learning Research (TMLR)*, 0, 2024.
3. Qian Huang, Jian Vora, Percy Liang, and Jure Leskovec. MLAGentBench: Evaluating language agents on machine learning experimentation. In *International Conference on Machine Learning (ICML)*, 2024.
4. Siddharth Karamcheti, Suraj Nair, Ashwin Balakrishna, Percy Liang, Thomas Kollar, and Dorsa Sadigh. Prismatic vlms: Investigating the design space of visually-conditioned language models. In *International Conference on Machine Learning (ICML)*, 2024.
5. Sayash Kapoor, Rishi Bommasani, Kevin Klyman, Shayne Longpre, Ashwin Ramaswami, Peter Cihon, Aspen Hopkins, Kevin Bankston, Stella Biderman, Miranda Bogen, Rumman Chowdhury, Alex Engler, Peter Henderson, Yacine Jernite, Seth Lazar, Stefano Maffulli, Alondra Nelson, Joelle Pineau, Aviya Skowron, Dawn Song, Victor Storch, Daniel Zhang, Daniel E. Ho, Percy Liang, and Arvind Narayanan. On the societal impact of open foundation models. In *International Conference on Machine Learning (Position Paper Track) (ICML)*, 2024.
6. Shayne Longpre, Sayash Kapoor, Kevin Klyman, Ashwin Ramaswami, Rishi Bommasani, Borhane Blili-Hamelin, Yangsibo Huang, Aviya Skowron, Zheng-Xin Yong, Suhas Kotha, Yi Zeng, Weiyan Shi, Xianjun Yang, Reid Southen, Alexander Robey, Patrick Chao, Diyi Yang, Ruoxi Jia, Daniel Kang, Sandy Pentland, Arvind Narayanan, Percy Liang, and Peter Henderson. A safe harbor for AI evaluation and red teaming. In *International Conference on Machine Learning (Position Paper Track) (ICML)*, 2024.
7. Xiang Lisa Li, Vaishnavi Shrivastava, Siyan Li, Tatsunori Hashimoto, and Percy Liang. Benchmarking and improving generator-validator consistency of language models. In *International Conference on Learning Representations (ICLR)*, 2024.
8. Michihiro Yasunaga, Xinyun Chen, Yujia Li, Panupong Pasupat, J. Leskovec, Percy Liang, Ed H. Chi, and Denny Zhou. Large language models as analogical reasoners. In *International Conference on Learning Representations (ICLR)*, 2024.
9. Chenchen Gu, Xiang Lisa Li, Percy Liang, and Tatsunori Hashimoto. On the learnability of watermarks for language models. In *International Conference on Learning Representations (ICLR)*, 2024.
10. S. Fleming, A. Lozano, W. Haberkorn, Jenelle A. Jindal, E. Reis, Rahul Thapa, L. Blankemeier, Julian Z. Jenkins, E. Steinberg, A. Nayak, Birju S. Patel, Chia-Chun Chiang, A. Callahan, Zepeng Huo, S. Gatidis, S. Adams, Oluseyi Fayanju, Shreya J. Shah, Thomas Savage, Ethan Goh, A. Chaudhari, N. Aghaeepour, Christopher D. Sharp, M. Pfeffer, Percy Liang, Jonathan H. Chen, K. Morse, E. Brunskill, Jason Alan Fries, and N. Shah. Medalign: A clinician-generated dataset for instruction following with electronic medical records. In *Association for the Advancement of Artificial Intelligence (AAAI)*, 2024.
11. Sang Michael Xie, Shibani Santurkar, Tengyu Ma, and Percy Liang. Data selection for language models via importance resampling. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
12. Sang Michael Xie, Hieu Pham, Xuanyi Dong, Nan Du, Hanxiao Liu, Yifeng Lu, Percy Liang, Quoc V. Le, Tengyu Ma, and A. Yu. Doremi: Optimizing data mixtures speeds up language model pretraining. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

13. Qian Huang, Hongyu Ren, Peng Chen, Gregor Krzyżanowicz, D. Zeng, Percy Liang, and J. Leskovec. PRODIGY: Enabling in-context learning over graphs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
14. Qian Huang, E. Zelikman, Sarah Chen, Yuhuai Wu, G. Valiant, and Percy Liang. Lexinvariant language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
15. Connor Toups, Rishi Bommasani, Kathleen A. Creel, Sarah H. Bana, Dan Jurafsky, and Percy Liang. Ecosystem-level analysis of deployed machine learning reveals homogeneous outcomes. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
16. Yann Dubois, Xuechen Li, Rohan Taori, Tianyi Zhang, Ishaan Gulrajani, Jimmy Ba, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. AlpacaFarm: A simulation framework for methods that learn from human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
17. Deepak Narayanan, Keshav Santhanam, Peter Henderson, Rishi Bommasani, Tony Lee, and Percy Liang. Cheaply evaluating inference efficiency metrics for autoregressive transformer apis. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
18. Tony Lee, Michihiro Yasunaga, Chenlin Meng, Yifan Mai, Joon Sung Park, Agrim Gupta, Yunzhi Zhang, Deepak Narayanan, Hannah Teufel, Marco Bellagente, Minguk Kang, Taesung Park, J. Leskovec, Jun-Yan Zhu, Fei-Fei Li, Jiajun Wu, Stefano Ermon, and Percy Liang. Holistic evaluation of text-to-image models. In *Neural Information Processing Systems Track on Datasets and Benchmarks (NeurIPS Datasets and Benchmarks)*, 2023.
19. Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics (TACL)*, 0, 2023.
20. Nelson F. Liu, Tianyi Zhang, and Percy Liang. Evaluating verifiability in generative search engines. In *Findings of Empirical Methods in Natural Language Processing (Findings of EMNLP)*, 2023.
21. Mina Lee, Megha Srivastava, Amelia Hardy, John Thickstun, Esin Durmus, Ashwin Paranjape, Ines Gerard-Ursin, Xiang Lisa Li, Faisal Ladhak, Frieda Rong, Rose E. Wang, Minae Kwon, Joon Sung Park, Hancheng Cao, Tony Lee, Rishi Bommasani, Michael Bernstein, and Percy Liang. Evaluating human-language model interaction. *Transactions of Machine Learning Research (TMLR)*, 0, 2023.
22. Rishi Bommasani and Percy Liang. Trustworthy social bias measurement. *arXiv*, 2023.
23. Tianyi Zhang, Faisal Ladhak, Esin Durmus, Percy Liang, K. McKeown, and Tatsunori Hashimoto. Benchmarking large language models for news summarization. *arXiv*, 2023.
24. Omar Khattab, Keshav Santhanam, Xiang Lisa Li, David Hall, Percy Liang, Christopher Potts, and Matei Zaharia. Demonstrate-Search-Predict: Composing retrieval and language models for knowledge-intensive NLP. *arXiv*, 2023.
25. Joon Sung Park, Joseph C. O’Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *User Interface Software and Technology (UIST)*, 2023.
26. Peter Henderson, Xuechen Li, Dan Jurafsky, Tatsunori Hashimoto, Mark A. Lemley, and Percy Liang. Foundation models and fair use. *Journal of Machine Learning Research (JMLR)*, 0, 2023.
27. Rishi Bommasani, Dilara Soylu, Thomas Liao, Kathleen A. Creel, and Percy Liang. Ecosystem graphs: The social footprint of foundation models. *arXiv*, 2023.

28. Siddharth Karamcheti, Suraj Nair, Annie S. Chen, T. Kollar, Chelsea Finn, Dorsa Sadigh, and Percy Liang. Language-driven representation learning for robotics. In *Robotics: Science and Systems (RSS)*, 2023.
29. Ying Sheng, Lianmin Zheng, Binhang Yuan, Zhuohan Li, Max Ryabinin, Daniel Y. Fu, Zhiqiang Xie, Beidi Chen, Clark W. Barrett, Joseph Gonzalez, Percy Liang, Christopher Ré, I. Stoica, and Ce Zhang. High-throughput generative inference of large language models with a single GPU. In *International Conference on Machine Learning (ICML)*, 2023.
30. Michihiro Yasunaga, Armen Aghajanyan, Weijia Shi, Rich James, Jure Leskovec, Percy Liang, Mike Lewis, Luke Zettlemoyer, and Wen tau Yih. Retrieval-augmented multimodal language modeling. In *International Conference on Machine Learning (ICML)*, 2023.
31. Shibani Santurkar, Esin Durmus, Faisal Ladhak, Cinoo Lee, Percy Liang, and Tatsunori Hashimoto. Whose opinions do language models reflect? In *International Conference on Machine Learning (ICML)*, 2023.
32. Yann Dubois, Tatsunori Hashimoto, and Percy Liang. Evaluating self-supervised learning via risk decomposition. In *International Conference on Machine Learning (ICML)*, 2023.
33. Irena Gao, Shiori Sagawa, Pang Wei Koh, Tatsunori Hashimoto, and Percy Liang. Out-of-domain robustness via targeted augmentations. In *International Conference on Machine Learning (ICML)*, 2023.
34. Nelson F. Liu, Ananya Kumar, Percy Liang, and Robin Jia. Are sample-efficient NLP models more robust? In *Association for Computational Linguistics (ACL)*, 2023.
35. Nelson F. Liu, Tony Lee, Robin Jia, and Percy Liang. Do question answering modeling improvements hold across benchmarks? In *Association for Computational Linguistics (ACL)*, 2023.
36. Yuhui Zhang, Michihiro Yasunaga, Zhengping Zhou, Jeff Z. HaoChen, James Zou, Percy Liang, and Serena Yeung. Beyond positive scaling: How negation impacts scaling trends of language models. In *Findings of the Association for Computational Linguistics (Findings of ACL)*, 2023.
37. Xiang Lisa Li, Ari Holtzman, Daniel Fried, Percy Liang, Jason Eisner, Tatsunori Hashimoto, Luke Zettlemoyer, and M. Lewis. Contrastive decoding: Open-ended text generation as optimization. In *Association for Computational Linguistics (ACL)*, 2023.
38. John Hewitt, John Thickstun, Christopher D. Manning, and Percy Liang. Backpack language models. In *Association for Computational Linguistics (ACL)*, 2023.
39. Yuchen Cui, Siddharth Karamcheti, Raj Pallethi, Nidhya Shivakumar, Percy Liang, and Dorsa Sadigh. ”no, to the right”– online language corrections for robotic manipulation via shared autonomy. In *ACM/IEEE International Conference on Human Robot Interaction (HRI)*, 2023.
40. Shibani Santurkar, Yann Dubois, Rohan Taori, Percy Liang, and Tatsunori Hashimoto. Is a caption worth a thousand images? a controlled study for representation learning. In *International Conference on Learning Representations (ICLR)*, 2023.
41. Yoonho Lee, Annie S. Chen, Fahim Tajwar, Ananya Kumar, Huaxiu Yao, Percy Liang, and Chelsea Finn. Surgical fine-tuning improves adaptation to distribution shifts. In *International Conference on Learning Representations (ICLR)*, 2023.
42. John Hewitt, Christopher D. Manning, and Percy Liang. Truncation sampling as language model desmoothing. In *Findings of Empirical Methods in Natural Language Processing (Findings of EMNLP)*, 2022.

43. Yuhuai Wu, Felix Li, and Percy Liang. Insights into pre-training via simpler synthetic tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
44. Yann Dubois, Tatsunori Hashimoto, S. Ermon, and Percy Liang. Improving self-supervised learning by characterizing idealized representations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
45. Binhang Yuan, Yongjun He, Jared Quincy Davis, Tianyi Zhang, Tri Dao, Beidi Chen, Percy Liang, Christopher Re, and Ce Zhang. Decentralized training of foundation models in heterogeneous environments. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
46. Xiang Lisa Li, John Thickstun, Ishaan Gulrajani, Percy Liang, and Tatsunori Hashimoto. Diffusion-LM improves controllable text generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
47. Shivam Garg, Dimitris Tsipras, Percy Liang, and G. Valiant. What can transformers learn in-context? a case study of simple function classes. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
48. Michihiro Yasunaga, Antoine Bosselut, Hongyu Ren, Xikun Zhang, Christopher D. Manning, Percy Liang, and Jure Leskovec. Deep bidirectional language-knowledge graph pretraining. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
49. Rishi Bommasani, Kathleen A. Creel, Ananya Kumar, Dan Jurafsky, and Percy Liang. Picking on the same person: Does algorithmic monoculture lead to outcome homogenization? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
50. Chris Donahue, John Thickstun, and Percy Liang. Melody transcription via generative pre-training. In *International Society for Music Information Retrieval (ISMIR)*, 2022.
51. J. Park, Lindsay Popowski, Carrie J. Cai, M. Morris, Percy Liang, and Michael S. Bernstein. Social simulacra: Creating populated prototypes for social computing systems. In *User Interface Software and Technology (UIST)*, 2022.
52. Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, Ed Chi, Tatsunori Hashimoto, Oriol Vinyals, Percy Liang, Jeff Dean, and William Fedus. Emergent abilities of large language models. *Transactions of Machine Learning Research (TMLR)*, 0, 2022.
53. Ananya Kumar, Tengyu Ma, Percy Liang, and Aditi Raghunathan. Calibrated ensembles can mitigate accuracy tradeoffs under distribution shift. In *Uncertainty in Artificial Intelligence (UAI)*, 2022.
54. Kendrick Shen, Robbie Jones, Ananya Kumar, Sang Michael Xie, Jeff Z. HaoChen, Tengyu Ma, and Percy Liang. Connect, not collapse: Explaining contrastive learning for unsupervised domain adaptation. In *International Conference on Machine Learning (ICML)*, 2022.
55. Michihiro Yasunaga, Jure Leskovec, and Percy Liang. LinkBERT: Pretraining language models with document links. In *Association for Computational Linguistics (ACL)*, 2022.
56. Shiori Sagawa, Pang Wei Koh, Tony Lee, Irena Gao, Sang Michael Xie, Kendrick Shen, Ananya Kumar, Weihua Hu, Michihiro Yasunaga, H. Marklund, Sara Beery, E. David, I. Stavness, Wei Guo, J. Leskovec, Kate Saenko, Tatsunori B. Hashimoto, S. Levine, Chelsea Finn, and Percy Liang. Extending the WILDS benchmark for unsupervised adaptation. In *International Conference on Learning Representations (ICLR)*, 2022.

57. Ananya Kumar, Aditi Raghunathan, Robbie Jones, Tengyu Ma, and Percy Liang. Fine-tuning can distort pretrained features and underperform out-of-distribution. In *International Conference on Learning Representations (ICLR)*, 2022.
58. Sang Michael Xie, Aditi Raghunathan, Percy Liang, and Tengyu Ma. An explanation of in-context learning as implicit Bayesian inference. In *International Conference on Learning Representations (ICLR)*, 2022.
59. Xuechen Li, Florian Tramèr, Percy Liang, and Tatsunori B. Hashimoto. Large language models can be strong differentially private learners. In *International Conference on Learning Representations (ICLR)*, 2022.
60. Xikun Zhang, Antoine Bosselut, Michihiro Yasunaga, Hongyu Ren, Percy Liang, Christopher D. Manning, and J. Leskovec. Greaselm: Graph reasoning enhanced language models for question answering. In *International Conference on Learning Representations (ICLR)*, 2022.
61. Mina Lee, Percy Liang, and Qian Yang. CoAuthor: Designing a human-AI collaborative writing dataset for exploring language model capabilities. In *Conference on Human Factors in Computing Systems (CHI)*, 2022.
62. Siddharth Karamcheti, Megha Srivastava, Percy Liang, and Dorsa Sadigh. LILA: Language-informed latent actions. In *Conference on Robot Learning (CoRL)*, 2021.
63. Michihiro Yasunaga, Jure Leskovec, and Percy Liang. LM-Critic: Language models for unsupervised grammatical error correction. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2021.
64. John Hewitt, Kawin Ethayarajh, Percy Liang, and Christopher D. Manning. Conditional probing: measuring usable information beyond a baseline. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2021.
65. Rodrigo Castellon, Chris Donahue, and Percy Liang. Codified audio language modeling learns useful representations for music information retrieval. In *International Society for Music Information Retrieval (ISMIR)*, 2021.
66. Michihiro Yasunaga and Percy Liang. Break-It-Fix-It: Unsupervised learning for program repair. In *International Conference on Machine Learning (ICML)*, 2021.
67. Jared Quincy Davis, Albert Gu, Krzysztof Choromanski, Tri Dao, Christopher Re, Chelsea Finn, and Percy Liang. Catformer: Designing stable transformers via sensitivity analysis. In *International Conference on Machine Learning (ICML)*, 2021.
68. Evan Zheran Liu, Behzad Haghgoo, Annie S. Chen, Aditi Raghunathan, Pang Wei Koh, Shiori Sagawa, Percy Liang, and Chelsea Finn. Just train twice: Improving group robustness without training group information. In *International Conference on Machine Learning (ICML)*, 2021.
69. Evan Zheran Liu, Aditi Raghunathan, Percy Liang, and Chelsea Finn. Decoupling exploration and exploitation for meta-reinforcement learning without sacrifices. In *International Conference on Machine Learning (ICML)*, 2021.
70. Sang Michael Xie, Tengyu Ma, and Percy Liang. Composed fine-tuning: Freezing pre-trained denoising autoencoders for improved generalization. In *International Conference on Machine Learning (ICML)*, 2021.
71. Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, Weihua Hu, Michihiro Yasunaga, Richard Lanus Phillips, Irena Gao, Tony Lee, Etienne David, Ian Stavness, Wei Guo, Berton A. Earnshaw, Imran S. Haque, Sara Beery, Jure Leskovec, Anshul Kundaje, Emma Pierson, Sergey Levine, Chelsea Finn, and Percy Liang. WILDS: A benchmark of in-the-wild distribution shifts. In *International Conference on Machine Learning (ICML)*, 2021.

72. John Miller, Rohan Taori, Aditi Raghunathan, Shiori Sagawa, Pang Wei Koh, Vaishaal Shankar, Percy Liang, Yair Carmon, and Ludwig Schmidt. Accuracy on the line: on the strong correlation between out-of-distribution and in-distribution generalization. In *International Conference on Machine Learning (ICML)*, 2021.
73. Mina Lee, C. Donahue, Robin Jia, Alexander Iyabor, and Percy Liang. Swords: A benchmark for lexical substitution with improved data coverage and quality. In *North American Association for Computational Linguistics (NAACL)*, 2021.
74. Michihiro Yasunaga, Hongyu Ren, Antoine Bosselut, Percy Liang, and Jure Leskovec. QA-GNN: Reasoning with language models and knowledge graphs for question answering. In *North American Association for Computational Linguistics (NAACL)*, 2021.
75. Xiang Lisa Li and Percy Liang. Prefix-tuning: Optimizing continuous prompts for generation. In *Association for Computational Linguistics (ACL)*, 2021.
76. Erik Jones, Shiori Sagawa, Pang Wei Koh, Ananya Kumar, and Percy Liang. Selective classification can magnify disparities across groups. In *International Conference on Learning Representations (ICLR)*, 2021.
77. Sang Michael Xie, Ananya Kumar, Robbie Jones, Fereshte Khani, Tengyu Ma, and Percy Liang. In-N-out: Pre-training and self-training using auxiliary information for out-of-distribution robustness. In *International Conference on Learning Representations (ICLR)*, 2021.
78. Fereshte Khani and Percy Liang. Removing spurious features can hurt accuracy and affect groups disproportionately. In *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2021.
79. Yu Gu, Sue Kase, Michelle T. Vanni, Brian M. Sadler, Percy Liang, Xifeng Yan, and Yu Su. Beyond i.i.d.: Three levels of generalization for question answering on knowledge bases. In *World Wide Web (WWW)*, 2021.
80. Pang Wei Koh, Jacob Steinhardt, and Percy Liang. Stronger data poisoning attacks break data sanitization defenses. *Machine Learning*, 1, 2021.
81. Stephen Mussmann, Robin Jia, and Percy Liang. On the importance of adaptive data collection for extremely imbalanced pairwise tasks. In *Findings of Empirical Methods in Natural Language Processing (Findings of EMNLP)*, 2020.
82. John Hewitt, Michael Hahn, Surya Ganguli, Percy Liang, and Christopher D. Manning. RNNs can generate bounded hierarchical languages with optimal memory. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2020.
83. Sumanth Dathathri, Krishnamurthy Dvijotham, Alexey Kurakin, Aditi Raghunathan, Jonathan Uesato, Rudy Bunel, Shreya Shankar, Jacob Steinhardt, Ian Goodfellow, Percy Liang, and Pushmeet Kohli. Enabling certification of verification-agnostic networks via memory-efficient semidefinite programming. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
84. Semantic Machines, Jacob Andreas, John Bufe, David Burkett, Charles Chen, Josh Clausman, Jean Crawford, Kate Crim, Jordan DeLoach, Leah Dorner, Jason Eisner, Hao Fang, Alan Guo, David Hall, Kristin Hayes, Kellie Hill, Diana Ho, Wendy Iwaszuk, Smriti Jha, Dan Klein, Jayant Krishnamurthy, Theo Lanman, Percy Liang, Christopher H. Lin, Ilya Lintsbakh, Andy McGovern, Aleksandr Nisnevich, Adam Pauls, Dmitriy Petters, Brent Read, Dan Roth, Subhro Roy, Jesse Rusak, Beth Short, Div Slomin, Ben Snyder, Stephon Striplin, Yu Su, Zachary Tellman, Sam Thomson, Andrei Vorobev, Izabela Witoszko, Jason Wolfe, Abby Wray, Yuchen Zhang, and Alexander Zotov. Task-oriented dialogue as dataflow synthesis. *Transactions of the Association for Computational Linguistics (TACL)*, 8, 2020.

85. Shiori Sagawa, Aditi Raghunathan, Pang Wei Koh, and Percy Liang. An investigation of why over-parameterization exacerbates spurious correlations. In *International Conference on Machine Learning (ICML)*, 2020.
86. Pang Wei Koh, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Pierson, Been Kim, and Percy Liang. Concept bottleneck models. In *International Conference on Machine Learning (ICML)*, 2020.
87. Fereshte Khani and Percy Liang. Feature noise induces loss discrepancy across groups. In *International Conference on Machine Learning (ICML)*, 2020.
88. Michihiro Yasunaga and Percy Liang. Graph-based, self-supervised program repair from diagnostic feedback. In *International Conference on Machine Learning (ICML)*, 2020.
89. Aditi Raghunathan, Sang Michael Xie, Fanny Yang, John C. Duchi, and Percy Liang. Understanding and mitigating the tradeoff between robustness and accuracy. In *International Conference on Machine Learning (ICML)*, 2020.
90. Ananya Kumar, Tengyu Ma, and Percy Liang. Understanding self-training for gradual domain adaptation. In *International Conference on Machine Learning (ICML)*, 2020.
91. Megha Srivastava, Tatsunori Hashimoto, and Percy Liang. Robustness to spurious correlations via human annotations. In *International Conference on Machine Learning (ICML)*, 2020.
92. Erik Jones, Robin Jia, Aditi Raghunathan, and Percy Liang. Robust encodings: A framework for combating adversarial typos. In *Association for Computational Linguistics (ACL)*, 2020.
93. Amita Kamath, Robin Jia, and Percy Liang. Selective question answering under domain shift. In *Association for Computational Linguistics (ACL)*, 2020.
94. Jesse Mu, Percy Liang, and Noah Goodman. Shaping visual representations with language for few-shot classification. In *Association for Computational Linguistics (ACL)*, 2020.
95. Shikhar Murty, Pang Wei Koh, and Percy Liang. ExpBERT: Representation engineering with natural language explanations. In *Association for Computational Linguistics (ACL)*, 2020.
96. Chris Donahue, Mina Lee, and Percy Liang. Enabling language models to fill in the blanks. In *Association for Computational Linguistics (ACL)*, 2020.
97. Shiori Sagawa, Pang Wei Koh, Tatsunori B. Hashimoto, and Percy Liang. Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. In *International Conference on Learning Representations (ICLR)*, 2020.
98. Weihua Hu, Bowen Liu, Joseph Gomes, Marinka Zitnik, Percy Liang, Vijay Pande, and Jure Leskovec. Strategies for pre-training graph neural networks. In *International Conference on Learning Representations (ICLR)*, 2020.
99. Cody Coleman, Christopher Yeh, Stephen Mussmann, Baharan Mirzasoleiman, Peter Bailis, Percy Liang, Jure Leskovec, and Matei Zaharia. Selection via proxy: Efficient data selection for deep learning. In *International Conference on Learning Representations (ICLR)*, 2020.
100. Ray Li, Percy Liang, and Stephen Mussmann. A tight analysis of greedy yields subexponential time approximation for uniform decision tree. In *Symposium on Discrete Algorithms (SODA)*, 2020.
101. Robin Jia, Aditi Raghunathan, Kerem Göksel, and Percy Liang. Certified robustness to adversarial word substitutions. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2019.

102. Yonatan Oren, Shiori Sagawa, Tatsunori Hashimoto, and Percy Liang. Distributionally robust language modeling. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2019.
103. John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2019.
104. Sumith Kulal, Panupong Pasupat, Kartik Chandra, Mina Lee, Oded Padon, Alex Aiken, and Percy Liang. Spoc: Search-based pseudocode to code. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
105. Yair Carmon, Aditi Raghunathan, Ludwig Schmidt, Percy Liang, and John C. Duchi. Unlabeled data improves adversarial robustness. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
106. Ananya Kumar, Percy Liang, and Tengyu Ma. Verified uncertainty calibration. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
107. Pang Wei Koh, Kai-Siang Ang, Hubert H. K. Teo, and Percy Liang. On the accuracy of influence functions for measuring group effects. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
108. Tatsunori Hashimoto, Hugh Zhang, and Percy Liang. Unifying human and statistical evaluation for natural language generation. In *North American Association for Computational Linguistics (NAACL)*, 2019.
109. Nanyun Peng, He He, and Percy Liang. Pun generation with surprise. In *North American Association for Computational Linguistics (NAACL)*, 2019.
110. Daniel Selsam, Matthew Lamm, Benedikt Bünz, Percy Liang, Leonardo de Moura, and David L. Dill. Learning a SAT solver from single-bit supervision. In *International Conference on Learning Representations (ICLR)*, 2019.
111. Yuchen Zhang and Percy Liang. Defending against whitebox adversarial attacks via randomized discretization. In *Artificial Intelligence and Statistics (AISTATS)*, 2019.
112. Emma Pierson, Pang Wei Koh, Tatsunori Hashimoto, Daphne Koller, Jure Leskovec, Nick Eriksson, and Percy Liang. Inferring multidimensional rates of aging from cross-sectional data. In *Artificial Intelligence and Statistics (AISTATS)*, 2019.
113. Kensen Shi, Jacob Steinhardt, and Percy Liang. FrAngel: Component-based synthesis with control structures. In *Principles of Programming Languages (POPL)*, 2019.
114. Aditi Raghunathan, Jacob Steinhardt, and Percy Liang. Semidefinite relaxations for certifying robustness to adversarial examples. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
115. Stephen Mussmann and Percy Liang. Uncertainty sampling is preconditioned stochastic gradient descent on zero-one loss. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
116. Tatsunori Hashimoto, Kelvin Guu, Yonatan Oren, and Percy Liang. A retrieve-and-edit framework for predicting structured outputs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
117. Eunsol Choi, He He, Mohit Iyyer, Mark Yatskar, Wen tau Yih, Yejin Choi, Percy Liang, and Luke Zettlemoyer. QuAC: Question answering in context. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2018.

118. He He, Derek Chen, Anusha Balakrishnan, and Percy Liang. Decoupling strategy and generation in negotiation dialogues. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2018.
119. Panupong Pasupat, Tian-Shun Jiang, Evan Zheran Liu, Kelvin Guu, and Percy Liang. Mapping natural language commands to web elements. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2018.
120. Matthew Lamm, Arun Chaganty, Christopher D. Manning, Dan Jurafsky, and Percy Liang. Textual analogy parsing: What’s shared and what’s compared among analogous facts. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2018.
121. Stephen Mussmann and Percy Liang. On the relationship between data efficiency and error in active learning. In *International Conference on Machine Learning (ICML)*, 2018.
122. Tatsunori B. Hashimoto, Megha Srivastava, Hongseok Namkoong, and Percy Liang. Fairness without demographics in repeated loss minimization. In *International Conference on Machine Learning (ICML)*, 2018.
123. Braden Hancock, Paroma Varma, Stephanie Wang, Martin Bringmann, Percy Liang, and Christopher Ré. Training classifiers with natural language explanations. In *Association for Computational Linguistics (ACL)*, 2018.
124. Arun Chaganty, Stephen Mussmann, and Percy Liang. The price of debiasing automatic metrics in natural language evaluation. In *Association for Computational Linguistics (ACL)*, 2018.
125. Pranav Rajpurkar, Robin Jia, and Percy Liang. Know what you don’t know: Unanswerable questions for SQuAD. In *Association for Computational Linguistics (ACL)*, 2018.
126. Stephen Mussmann and Percy Liang. Generalized binary search for split-neighborly problems. In *Artificial Intelligence and Statistics (AISTATS)*, 2018.
127. Fereshte Khani, Noah D. Goodman, and Percy Liang. Planning, inference and pragmatics in sequential language games. *Transactions of the Association for Computational Linguistics (TACL)*, 6, 2018.
128. Kelvin Guu, Tatsunori B. Hashimoto, Yonatan Oren, and Percy Liang. Generating sentences by editing prototypes. *Transactions of the Association for Computational Linguistics (TACL)*, 0, 2018.
129. Juncen Li, Robin Jia, He He, and Percy Liang. Delete, retrieve, generate: A simple approach to sentiment and style transfer. In *North American Association for Computational Linguistics (NAACL)*, 2018.
130. Evan Zheran Liu, Kelvin Guu, Panupong Pasupat, Tianlin Shi, and Percy Liang. Reinforcement learning on web interfaces using workflow-guided exploration. In *International Conference on Learning Representations (ICLR)*, 2018.
131. Aditi Raghunathan, Jacob Steinhardt, and Percy Liang. Certified defenses against adversarial examples. In *International Conference on Learning Representations (ICLR)*, 2018.
132. Osbert Bastani, Rahul Sharma, Alex Aiken, and Percy Liang. Active learning of points-to specifications. In *Programming Language Design and Implementation (PLDI)*, 2018.
133. Vatsal Sharan, Sham Kakade, Percy Liang, and Gregory Valiant. Prediction with a short memory. In *Symposium on Theory of Computing (STOC)*, 2018.
134. Jacob Steinhardt, Pang Wei Koh, and Percy Liang. Certified defenses for data poisoning attacks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

135. Tatsunori B. Hashimoto, John Duchi, and Percy Liang. Unsupervised transformation learning via convex relaxations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
136. Vatsal Sharan, Sham Kakade, Percy Liang, and Gregory Valiant. Learning overcomplete HMMs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
137. Robin Jia and Percy Liang. Adversarial examples for evaluating reading comprehension systems. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2017.
138. Yuchen Zhang, Panupong Pasupat, and Percy Liang. Macro grammars and holistic triggering for efficient semantic parsing. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2017.
139. Arun Chaganty, Ashwin Paranjape, Percy Liang, and Chris Manning. Importance sampling for unbiased on-demand evaluation of knowledge base population. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2017.
140. Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions. In *International Conference on Machine Learning (ICML)*, 2017.
141. Yuchen Zhang, Percy Liang, and Martin J. Wainwright. Convexified convolutional neural networks. In *International Conference on Machine Learning (ICML)*, 2017.
142. Daniel Selsam, Percy Liang, and David Dill. Developing bug-free machine learning systems with formal mathematics. In *International Conference on Machine Learning (ICML)*, 2017.
143. Tianlin Shi, Andrej Karpathy, Linxi Fan, Jonathan Hernandez, and Percy Liang. World of bits: An open-domain platform for web-based agents. In *International Conference on Machine Learning (ICML)*, 2017.
144. Yuchen Zhang, Percy Liang, and Moses Charikar. A hitting time analysis of stochastic gradient Langevin dynamics. In *Conference on Learning Theory (COLT)*, 2017.
145. Osbert Bastani, Rahul Sharma, Alex Aiken, and Percy Liang. Synthesizing program input grammars. In *Programming Language Design and Implementation (PLDI)*, 2017.
146. Sida I. Wang, Sam Ginn, Percy Liang, and Christopher D. Manning. Naturalizing a programming language via interactive learning. In *Association for Computational Linguistics (ACL)*, 2017.
147. He He, Anusha Balakrishnan, Mihail Eric, and Percy Liang. Learning symmetric collaborative dialogue agents with dynamic knowledge graph embeddings. In *Association for Computational Linguistics (ACL)*, pages 1766–1776, 2017.
148. Kelvin Guu, Panupong Pasupat, Evan Zheran Liu, and Percy Liang. From language to programs: Bridging reinforcement learning and maximum marginal likelihood. In *Association for Computational Linguistics (ACL)*, 2017.
149. Jacob Steinhardt and Percy Liang. Unsupervised risk estimation using only conditional independence structure. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.
150. Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. SQuAD: 100,000+ questions for machine comprehension of text. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2016.
151. Sida I. Wang, Percy Liang, and Chris Manning. Learning language games through interaction. In *Association for Computational Linguistics (ACL)*, 2016. Acceptance rate: 25.4% (328/1290).
152. Robin Jia and Percy Liang. Data recombination for neural semantic parsing. In *Association for Computational Linguistics (ACL)*, 2016. Acceptance rate: 25.4% (328/1290).

153. Reginald Long, Panupong Pasupat, and Percy Liang. Simpler context-dependent logical forms via model projections. In *Association for Computational Linguistics (ACL)*, 2016. Acceptance rate: 25.4% (328/1290).
154. Panupong Pasupat and Percy Liang. Inferring logical forms from denotations. In *Association for Computational Linguistics (ACL)*, 2016. Acceptance rate: 25.4% (328/1290).
155. Fereshte Khani, Martin Rinard, and Percy Liang. Unanimous prediction for 100% precision with application to learning semantic mappings. In *Association for Computational Linguistics (ACL)*, 2016. Acceptance rate: 25.4% (328/1290).
156. Arun Tejasvi Chaganty and Percy Liang. How much is 131 million dollars? Putting numbers in perspective with compositional descriptions. In *Association for Computational Linguistics (ACL)*, 2016. Acceptance rate: 25.4% (328/1290).
157. Aditi Raghunathan, Roy Frostig, John Duchi, and Percy Liang. Estimation from indirect supervision with linear moments. In *International Conference on Machine Learning (ICML)*, 2016.
158. Yushi Wang, Jonathan Berant, and Percy Liang. Building a semantic parser overnight. In *Association for Computational Linguistics (ACL)*, 2015. Acceptance rate: 25% (173/692).
159. Jonathan Berant and Percy Liang. Imitation learning of agenda-based semantic parsers. *Transactions of the Association for Computational Linguistics (TACL)*, 3:545–558, 2015.
160. Jacob Steinhardt and Percy Liang. Learning with relaxed supervision. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.
161. Sida I. Wang, Arun Chaganty, and Percy Liang. Estimating mixture models via mixture of polynomials. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.
162. Keenon Werling, Arun Chaganty, Percy Liang, and Chris Manning. On-the-job learning with Bayesian decision theory. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.
163. Volodymyr Kuleshov and Percy Liang. Calibrated structured prediction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.
164. Volodymyr Kuleshov, Arun Chaganty, and Percy Liang. Simultaneous diagonalization: the asymmetric, low-rank, and noisy settings. *arXiv*, 2015.
165. Kelvin Guu, John Miller, and Percy Liang. Traversing knowledge graphs in vector space. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2015. Acceptance rate: 24% (312/1315).
166. Panupong Pasupat and Percy Liang. Compositional semantic parsing on semi-structured tables. In *Association for Computational Linguistics (ACL)*, 2015. Acceptance rate: 25% (173/692).
167. Dipendra K. Misra, Kejia Tao, Percy Liang, and Ashutosh Saxena. Environment-driven lexicon induction for high-level instructions. In *Association for Computational Linguistics (ACL)*, 2015. Acceptance rate: 25% (173/692).
168. Jacob Steinhardt and Percy Liang. Reified context models. In *International Conference on Machine Learning (ICML)*, 2015. Acceptance rate: 26%.
169. Jacob Steinhardt and Percy Liang. Learning fast-mixing models for structured prediction. In *International Conference on Machine Learning (ICML)*, pages 1063–1072, 2015. Acceptance rate: 26%.
170. Tianlin Shi, Jacob Steinhardt, and Percy Liang. Learning where to sample in structured prediction. In *Artificial Intelligence and Statistics (AISTATS)*, pages 875–884, 2015. Acceptance rate: 28.7%.

171. Volodymyr Kuleshov, Arun Chaganty, and Percy Liang. Tensor factorization via matrix factorization. In *Artificial Intelligence and Statistics (AISTATS)*, 2015. Acceptance rate: 28.7%.
172. Vignesh Ramanathan, Armand Joulin, Percy Liang, and Li Fei-Fei. Linking people with "their" names using coreference resolution. In *European Conference on Computer Vision (ECCV)*, 2014.
173. Jonathan Berant and Percy Liang. Semantic parsing via paraphrasing. In *Association for Computational Linguistics (ACL)*, 2014. Acceptance rate: 26.2% (146/572).
174. Panupong Pasupat and Percy Liang. Zero-shot entity extraction from web pages. In *Association for Computational Linguistics (ACL)*, 2014. Acceptance rate: 26.2% (146/572).
175. Arun Chaganty and Percy Liang. Estimating latent-variable graphical models using moments and likelihoods. In *International Conference on Machine Learning (ICML)*, 2014. Acceptance rate: 22%.
176. Jacob Steinhardt and Percy Liang. Adaptivity and optimism: An improved exponentiated gradient algorithm. In *International Conference on Machine Learning (ICML)*, 2014. Acceptance rate: 22%.
177. Jacob Steinhardt and Percy Liang. Filtering with abstract particles. In *International Conference on Machine Learning (ICML)*, pages 727–735, 2014. Acceptance rate: 22%.
178. Stefan Wager, Will Fithian, Sida I. Wang, and Percy Liang. Altitude training: Strong bounds for single-layer dropout. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.
179. Roy Frostig, Sida I. Wang, Percy Liang, and Chris Manning. Simple MAP inference via low-rank relaxations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.
180. Sida I. Wang, Roy Frostig, Percy Liang, and Chris Manning. Relaxations for inference in restricted Boltzmann machines. In *International Conference on Learning Representations Workshop (ICLR)*, 2014.
181. Percy Liang. Lambda dependency-based compositional semantics. *arXiv preprint arXiv:1309.4408*, 2013.
182. Jonathan Berant, Andrew Chou, Roy Frostig, and Percy Liang. Semantic parsing on Freebase from question-answer pairs. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2013. Acceptance rate: 27% (205/772).
183. Sida I. Wang, Mengqiu Wang, Stefan Wager, Percy Liang, and Chris Manning. Feature noising for log-linear structured prediction. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2013. Acceptance rate: 27% (205/772).
184. Stefan Wager, Sida I. Wang, and Percy Liang. Dropout training as adaptive regularization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2013.
185. Arun Chaganty and Percy Liang. Spectral experts for estimating mixtures of linear regressions. In *International Conference on Machine Learning (ICML)*, 2013. Acceptance rate: 24% (283/1204).
186. Vignesh Ramanathan, Percy Liang, and Li Fei-Fei. Video event understanding using natural language descriptions. In *International Conference on Computer Vision (ICCV)*, 2013.
187. Rahul Sharma, Saurabh Gupta, Bharath Hariharan, Alex Aiken, Percy Liang, and Aditya V. Nori. A data driven approach for algebraic loop invariants. In *European Symposium on Programming (ESOP)*, 2013.
188. Daniel Hsu, Sham M. Kakade, and Percy Liang. Identifiability and unmixing of latent parse trees. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2012.

189. Percy Liang, Michael I. Jordan, and Dan Klein. Learning dependency-based compositional semantics. In *Association for Computational Linguistics (ACL)*, pages 590–599, 2011. Acceptance rate: 26% (164/634).
190. Percy Liang and Mayur Naik. Scaling up abstraction refinement via pruning. In *Programming Language Design and Implementation (PLDI)*, 2011.
191. Percy Liang, Omer Tripp, and Mayur Naik. Learning minimal abstractions. In *Principles of Programming Languages (POPL)*, 2011.
192. Percy Liang, Michael I. Jordan, and Dan Klein. Learning programs: A hierarchical Bayesian approach. In *International Conference on Machine Learning (ICML)*, pages 639–646, 2010. Acceptance rate: 25.9% (152/594).
193. Percy Liang and Nati Srebro. On the interaction between norm and dimensionality: Multiple regimes in learning. In *International Conference on Machine Learning (ICML)*, 2010. Acceptance rate: 25.9% (152/594).
194. Percy Liang, Michael I. Jordan, and Dan Klein. Type-based MCMC. In *North American Association for Computational Linguistics (NAACL)*, 2010. Acceptance rate: 31% (90/291).
195. Percy Liang, Omer Tripp, Mayur Naik, and Mooly Sagiv. A dynamic evaluation of static heap abstractions. In *Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*, 2010.
196. Dave Golland, Percy Liang, and Dan Klein. A game-theoretic approach to generating spatial descriptions. In *Empirical Methods in Natural Language Processing (EMNLP)*, pages 410–419, 2010. Acceptance rate: 25% (125/500).
197. Gabor Angeli, Percy Liang, and Dan Klein. A simple domain-independent probabilistic approach to generation. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2010. Acceptance rate: 25% (125/500).
198. Percy Liang, Francis Bach, Guillaume Bouchard, and Michael I. Jordan. Asymptotically optimal regularization in smooth parametric models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2009.
199. Percy Liang, Michael I. Jordan, and Dan Klein. Learning semantic correspondences with less supervision. In *Association for Computational Linguistics and International Joint Conference on Natural Language Processing (ACL-IJCNLP)*, pages 91–99, 2009.
200. Percy Liang and Dan Klein. Online EM for unsupervised models. In *North American Association for Computational Linguistics (NAACL)*, pages 611–619, 2009.
201. Percy Liang, Michael I. Jordan, and Dan Klein. Learning from measurements in exponential families. In *International Conference on Machine Learning (ICML)*, 2009. Acceptance rate: 26.9% (160/595).
202. Percy Liang, Dan Klein, and Michael I. Jordan. Agreement-based learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2008.
203. Percy Liang and Michael I. Jordan. An asymptotic analysis of generative, discriminative, and pseudolikelihood estimators. In *International Conference on Machine Learning (ICML)*, pages 584–591, 2008. Acceptance rate: 27% (155/583).
204. Percy Liang, Hal Daumé III, and Dan Klein. Structure compilation: Trading structure for features. In *International Conference on Machine Learning (ICML)*, 2008. Acceptance rate: 27% (155/583).

205. Percy Liang and Dan Klein. Analyzing the errors of unsupervised learning. In *Human Language Technology and Association for Computational Linguistics (HLT/ACL)*, 2008.
206. Aria Haghighi, Percy Liang, Taylor Berg-Kirkpatrick, and Dan Klein. Learning bilingual lexicons from monolingual corpora. In *Human Language Technology and Association for Computational Linguistics (HLT/ACL)*, 2008.
207. Alexandre Bouchard-Côté, Percy Liang, Tom Griffiths, and Dan Klein. A probabilistic approach to language change. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2008.
208. Percy Liang, Michael I. Jordan, and Ben Taskar. A permutation-augmented sampler for Dirichlet process mixture models. In *International Conference on Machine Learning (ICML)*, 2007. Acceptance rate: 29% (150/522).
209. Alexandre Bouchard-Côté, Percy Liang, Tom Griffiths, and Dan Klein. A probabilistic approach to diachronic phonology. In *Empirical Methods in Natural Language Processing and Computational Natural Language Learning (EMNLP/CoNLL)*, 2007.
210. Percy Liang, Slav Petrov, Michael I. Jordan, and Dan Klein. The infinite PCFG using hierarchical Dirichlet processes. In *Empirical Methods in Natural Language Processing and Computational Natural Language Learning (EMNLP/CoNLL)*, 2007.
211. Percy Liang, Alexandre Bouchard-Côté, Dan Klein, and Ben Taskar. An end-to-end discriminative approach to machine translation. In *International Conference on Computational Linguistics and Association for Computational Linguistics (COLING/ACL)*, 2006.
212. Percy Liang, Ben Taskar, and Dan Klein. Alignment by agreement. In *North American Association for Computational Linguistics (NAACL)*, pages 104–111, 2006.
213. Percy Liang, Mukund Narasimhan, Michael Shilman, and Paul Viola. Efficient geometric algorithms for parsing in two dimensions. In *International Conference on Document Analysis and Recognition (ICDAR)*, 2005.

Professional Activities

Reviewer for conferences International Conference on Machine Learning (ICML), Computational Learning Theory (COLT), Neural Information Processing Systems (NeurIPS), North American Association of Computational Linguistics (NAACL), Association of Computational Linguistics (ACL), Empirical Methods in Natural Language Processing (EMNLP), Uncertainty in Artificial Intelligence (UAI), Artificial Intelligence and Statistics (AISTATS), International Joint Conference on Artificial Intelligence (IJCAI), Principles of Programming Languages (POPL).

Reviewer for journals Transactions of Association of Computational Linguistics (TACL), Journal of Artificial Intelligence Research (JAIR), Journal of Machine Learning Research (JMLR), IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI).

Area chair for conferences North American Association of Computational Linguistics (NAACL), International Conference on Machine Learning (ICML), Association of Computational Linguistics (ACL).

Program chair for conferences Neural Information Processing Systems (NeurIPS).

Teaching

2012–2023	Artificial Intelligence: Principles and Techniques (Stanford CS221)
2013–2016	Statistical Learning Theory (Stanford CS229T)
2021–2022	Advances in Foundation Models (Stanford CS324)
2023	Language Models From Scratch (Stanford CS336)
2008	Introduction to Artificial Intelligence (Berkeley, teaching assistant)
2007	Statistical Learning Theory (CS281A) (Berkeley, teaching assistant)